

# LGU Gingoog Agriculture Management System

## Final System Process Flow

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Prepared for: LGU Gingoog - Office of Agriculture

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### Overview

The LGU Gingoog Agriculture Management System is a role-based web platform for managing agriculture and fishery records, location references, resources, employee information, field check-ins, analytics, and reports.

The final system flow follows this sequence:

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System Access

Administrative Setup

Registry Enrollment

Location and Risk Setup

Sector Operations

Resource and Finance Management

Employee Operations

Dashboard Monitoring

Report Generation

““

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### Phase 1: System Access

““

User opens the system URL

Login Page

Enter username/email and password

System validates account and role

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### Login Decision

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Valid credentials + valid role

Redirect to Dashboard

Valid credentials but no assigned role

Redirect to No Role page

Invalid credentials  
Show login error  
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## Password Recovery

""  
Click "Forgot Password?"  
Enter registered email  
Receive password reset link  
Open reset page  
Set new password  
Return to Login  
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## Authenticated Access

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Authenticated user opens a page  
System checks required page permission  
If allowed: display page  
If not allowed: display Page Not Available  
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## Phase 2: Administrative Setup

This baseline setup is normally performed by the Administrator before regular daily operations.

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Administrator logs in  
Configure Role Management  
Configure User Management  
Configure system reference data  
Register employee profiles  
Assign permissions and module access  
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## Role Management

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Create role  
Select module permissions  
Save role  
Role becomes available for user assignment

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Permission groups include:

““

Dashboard

Overview analytics

Farmer Registry

View/manage registered farmers

Fisherfolk Registry

View/manage registered fisherfolk

Cooperatives

View/manage FFCA cooperative records

Locations

Barangay list, work locations, danger zones

Production

Crops, planting logs, harvest records

Fishery

Fishery production records

Resources

Inventory and equipment records

Finance

Expenses and financial reports

Administration

Employees and employee logs

Access Control

Roles and users

System Settings

Global settings access

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## User Management

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Create user account

Link account to employee record where applicable

Assign role

Save account  
User can log in based on assigned role permissions

### Phase 3: Registry Enrollment

Registry records serve as the foundation for production, fishery, finance, barangay, and reporting modules.

Farmer Registry  
Register farmer personal information  
Encode farm profile and farm location  
Link barangay, crop, and cooperative where applicable  
Save farmer record

Fisherfolk Registry  
Register fisherfolk personal information  
Encode fishery profile  
Link barangay and cooperative where applicable  
Save fisherfolk record

FFCA / Cooperatives  
Register cooperative profile  
Link farmer and fisherfolk members  
Use membership data in barangay and registry views

### Registry Dependency Rule

Farmer records should exist before planting, harvest, and farmer-linked expenses.  
Fisherfolk records should exist before fishery production records.  
Cooperative records should exist before assigning members to FFCA groups.

### Phase 4: Location and Risk Management

Location modules support registry classification, mapping, field assignments, and risk overlays.

#### Barangay Profile

- Open Barangay Profile
  - View barangay metrics
  - Review linked farmers, fisherfolk, cooperatives, crops, and planting logs
  - Open maps and profile details
  - Edit barangay information when permitted

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## Work Location

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- Open Work Location
  - Add cluster/department/work-location record
  - Set name, description, and status
  - Save record
  - Use as employee assignment reference

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## Danger Zones

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- Open Danger Zones
  - Add danger zone
  - Enter zone name, type, status, colors, and description
  - Add polygon coordinates manually or upload GPX
  - Preview polygon on map
  - Save danger zone

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Danger zone polygons are used as map-based risk references for location-aware workflows.

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## Phase 5: Sector Operations

### Crop Agriculture

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- Crops
  - Maintain crop reference records
  - Use crops in farmer and planting records
- Planting Logs
  - Select registered farmer
  - Select farm/location and crop
  - Encode planting date, area, status, and remarks
  - Save planting log

## Harvest Records

- Link harvest to planting/farmer/crop context
- Encode harvest date, quantity/yield, quality, and status
- Save harvest record

## Fishery

### Fisheries

- Select registered fisherfolk/operator
- Encode fishery type/facility, production volume, date, and remarks
- Save fishery record
- Linked data updates fisherfolk and reporting views

## Phase 6: Resource and Finance Management

### Inventory

#### Add inventory item

- Encode item details and unit
- Save item
- Record stock-in or stock-out transaction
- Review inventory balances and transaction logs

### Equipments

#### Register equipment

- Encode equipment identity, category, condition, status, and location
- Save equipment
- View, update, or delete equipment record when permitted

### Expenses

#### Log expense

- Select category
- Enter amount, date, description, and optional farmer link
- Save expense
- Monitor expense table and monthly cost breakdown

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## Phase 7: Employee Operations

### Employee Information

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Open Employee Information

- Add employee profile

- Encode employee number, name, position, department, work location, supervisor, status, and employment type

- Upload optional face reference image

- Save employee record

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Employee records support user-account linkage, hierarchy views, and Smart Check-In verification.

### Employee Logs / Smart Check-In

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Employee opens Employee Logs

- Click Smart Check-In

- Select employee profile

- For non-admin users, profile is locked to the linked employee

- Turn on camera

- Capture face image

- Verify face against employee reference image

- Capture GPS location

- Save check-in log with timestamp, coordinates, place, status, and match score

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Access behavior:

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Administrator / permitted admin role

- Can view employee logs according to permissions

Non-admin employee

- Can view only own logs when linked to an employee profile

User without permission

- Cannot open Employee Logs page

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## Phase 8: Dashboard Monitoring

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Dashboard opens after login  
Load key metrics  
Display charts and map summaries  
Show recent activities  
Show weather and barangay weather context  
Provide quick actions

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Quick actions include:

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Add Planting Log  
Planting Logs

Register Farmer  
Farmer Registry

Register Fisherfolk  
Fisherfolk Registry

Log Expense  
Expenses

““

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## Phase 9: Reports

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Open Reports  
Click Generate Report  
Select report category and data module  
Select fields, filters, date range, and output format  
Generate report  
View full preview  
Download PDF or XLSX output

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Report categories include census, production, barangay profile, financial, and related module-based reports.

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## Final Overall Data Flow



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#### Roles and Users

Control module access

#### Barangay Profiles + Work Locations + Danger Zones

Support registry, employee, map, and risk references

#### Cooperatives

Link farmers and fisherfolk

#### Farmers

Link to crops, farm locations, planting logs, harvest records, expenses, barangays, and reports

#### Fisherfolk

Link to fishery records, cooperatives, barangays, and reports

#### Crops

Link to farmer profiles, planting logs, harvest records, dashboard analytics, and reports

#### Inventory + Equipments + Expenses

Support resource tracking, finance monitoring, and reporting

#### Employees

Link to users, work locations, supervisors, and employee logs

#### All active modules

Feed dashboard analytics and report generation

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## Active Module List

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#### 1. Dashboard

Overview metrics, charts, weather, map summaries, and quick actions

#### 2. Farmer Registry

Register and manage farmer profiles and farm details

#### 3. Fisherfolk Registry

Register and manage fisherfolk profiles and fishery details

#### 4. FFCA (Cooperatives)

Manage cooperative profiles and member links

5. Barangay Profile  
View barangay-level records, tabs, metrics, and maps
6. Work Location  
Manage work-location reference records
7. Danger Zones  
Manage polygon-based risk areas
8. Crops  
Manage crop reference records
9. Planting Logs  
Track planting activities
10. Harvest Records  
Track harvest outputs
11. Fisheries  
Track fishery production records
12. Inventory  
Manage stock items and stock transactions
13. Equipments  
Manage equipment records and status
14. Expenses  
Track agricultural expenses
15. Reports  
Generate, preview, and download reports
16. Employee Information  
Manage employee directory and hierarchy
17. Employee Logs  
Record Smart Check-In logs with face and GPS verification
18. Role Management  
Manage role permissions
19. User Management  
Manage user accounts and role assignment
20. Settings  
Manage profile, password, and appearance preferences

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## Final Access Summary

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Administrator

Full system access

Staff / Encoder

Access based on assigned view/manage permissions

Viewer / Limited User

Read-only or restricted access based on role permissions

Employee-linked User

Can access own employee-related records where allowed

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\*LGU Gingoog Agriculture Management System - Final System Process Flow v2.0\*

\*Office of Agriculture, Gingoog City, Misamis Oriental\*